Royal Devon University Healthcare NHS Foundation Trust

Above Cuff Vocalisation for Tracheostomy patients in Intensive Care SOP, Northern Services

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Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	**Yes**
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:			

Note: This document has been assessed for any equality, diversity or human rights implications

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CONTENTS

1. Introduction 4

2. Purpose 4

3. Definitions 4

4. Duties and Responsibilities of Staff 4

5. Main Body 5
Suitable patients **5**
Contraindications **5**
Equipment required for trial of ACV **5**
Procedure for trial of ACV..... **6**
Potential Side effects **10**
Documentation **10**

6. Archiving arrangements 10

7. Process for Monitoring and Compliance 10

8. References 11

9. Appendix 1 Communication Plan 12

1. INTRODUCTION

Cuffed tracheostomy tubes prevent airflow through the larynx, preventing vocalisation. For some patients, it is not possible to deflate the tracheostomy tube cuff due to high pressures on the ventilator or high subglottic aspirates. Above Cuff Vocalisation (ACV) via a tracheostomy with a subglottic port can help in these situations to facilitate laryngeal rehabilitation and aid communication for patients.

This technique uses the subglottic suction port of specialised tracheostomy tubes to deliver a low flow of gas (air or oxygen) backwards, up the subglottic suction port, to exit above the cuff. This gas flow can then travel upwards through the trachea, pass through the vocal cords and exit via the mouth. This can result in audible vocalisation in around 80% of patients who would otherwise not be able to speak.

Sub-glottic suction tubes, for example the 'Trachoe-Twist Plus' and 'Blue Line Ultra Suction Aid' are used primarily to remove sub-glottic secretions, but retrograde gas air flows via the suction port can facilitate above cuff vocalisation (ACV).

2. PURPOSE

To outline the safe process for use of tracheostomy above cuff vocalisation (ACV) for patients with a tracheostomy with a subglottic port within ICU in Northern services.

3. DEFINITIONS

Tracheostomy: an opening in the front of the neck, into the trachea so a tube can be inserted to facilitate breathing.

Sub glottic port: a small hole in the shaft of the tracheostomy tube just above the cuff.

ACV: Above Cuff Vocalisation

ICU: Intensive Care Unit

SLT: Speech and Language Therapy

4. DUTIES AND RESPONSIBILITIES OF STAFF

Due to the risks associated with this procedure it should be carried out with ICU consultant's agreement.

The procedure should only be carried out by staff trained by a consultant, or the leads of Speech and Language therapy and Physiotherapists trained in ACV.

5. MAIN BODY

Suitable patients

- Stable on current ventilation settings.
- Awake
- No upper airway obstruction / abnormality
- Healthy tracheostomy stoma, more than 48-72 hours post insertion
- Early cuff deflation ruled out, or, transitioning to cuff deflation
- Sub glottic suction tube in situ, cuff inflated
- Can be mechanically ventilated or self-ventilating with a tracheostomy tube.

Contraindications

- A new tracheostomy insertion less 48-72 hours
- Laryngectomy
- Upper airway obstruction or concerns about patency of the airway
- An altered upper airway
- Any problems with the stoma such as infection, bleeding or tissue breakdown

Equipment Required for trial of ACV:

- 10ml syringe for subglottic suction port
- Oxygen tubing
- Clear open valve. Figure 1 shows valves provided with tracheostomy tube, figure 2 shows valve available to order as alternative option.
- Oxygen cylinder

Figure 1

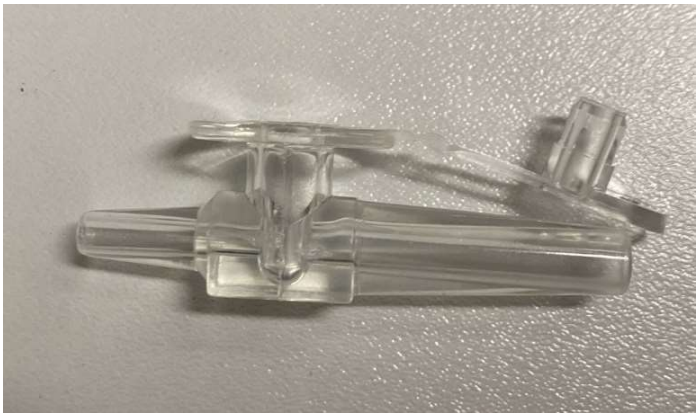


Figure 2

Procedure for trial of ACV:

Action	Rationale
Discuss with Multi-disciplinary team and Consultant Intensivist. Discuss agreed plan with nurse in charge and nurse in	To ensure all contraindications are discussed and all members of the team are aware of the plan.

bedspace. Consultant to be on the unit for the first session and agree it's a convenient time in case of support needed.	Some adverse events documented under potential side effects from the literature.
Check for contraindications	To ensure safety
Explain procedure to the patient and gain consent	To gain patient consent and minimise anxiety
Sit patient upright (unless contra-indicated e.g., spinal cord injury)	To optimise lung expansion
Assess patients' respiratory status i.e., rate, depth, and pattern	Ensures appropriate and that the patient remains stable
Ensure the patient is continually monitored during ACV and oral suction set up.	Safety. Patients will frequently cough when commencing ACV as the larynx regains sensation.
Cut oxygen tubing at correct points and connect oxygen tubing to the oxygen port so it's ready.	
Wash hands and don apron and non-sterile gloves	Infection control
Use a syringe on the sub-glottic port (figure 3 shows subglottic port) to withdraw and remove any secretions sitting on top of the cuff. Use yankeur suction in mouth. Warn the patient that they can expect some secretions to be 'blown' into their mouth once the ACV begins. Be aware this could be unpleasant.	Remove excess sub-glottic secretions and any oral secretions. These may prevent vocalisation.
Connect from the oxygen port to the sub-glottic suction port, via the clear open valve (figure 4).	Delivers oxygen via sub-glottic suction port with valve ensuring gas flow only when required. Using a thumb port

Use a 'thumb port' so that there is only continuous gas flow when the port is actively occluded, demonstrated in figure 5	minimises the risk of a pressure build up should the ACV port become obstructed.
Turn oxygen flow rate to between 2 and 5 litres per minute	Varying flow rates required for individual patients, start low on 2 litres per minute and increase as needed only. Larynx has not been used for some time and can be sensitive.
Occlude clear open valve shown in Figure 6 when ready to trial voice	Allow retrograde laryngeal airflow to facilitate voice
Encourage patient phonation while valve occluded.	To promote phonation
Do not leave the patient during the procedure	Safety
Limit use of ACV to no more than 15 minutes. The amount of times completed is patient dependent. Normally this is 1-3 times a day but if particularly helpful and pivotal part of weaning it can be completed with 2 hour gaps in the day. The more frequent it is performed the higher the risk of laryngeal drying so monitor for this and increase frequency over a number of days.	Prevent potential of laryngeal drying. Safety
Remove oxygen tubing and clear open valve. Provide suction and subglottic suction as required	Remove excessive secretions present
Remove gloves and aprons. Wash hands with soap and water.	

Stop the trial if there is any evidence of:

- Gas escaping via the stoma, around the tracheostomy tube
- Pain or discomfort reported by the patient
- Swelling or surgical emphysema that develops at the site
- The patient becomes fatigued / sleepy

Figure 3

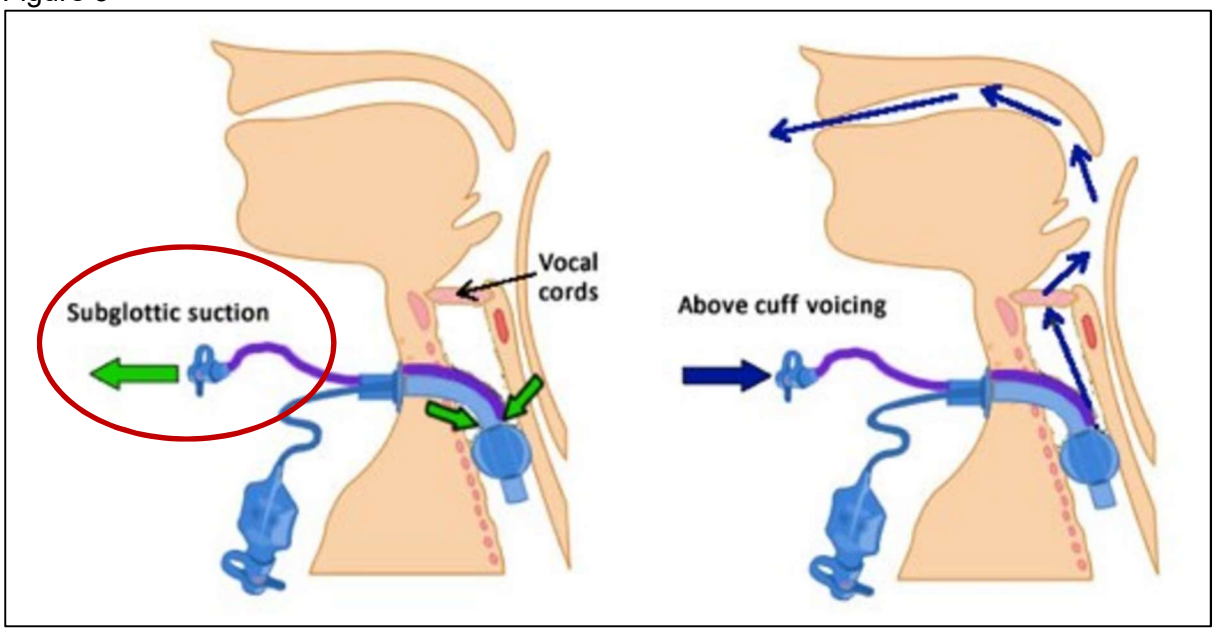


Figure 4

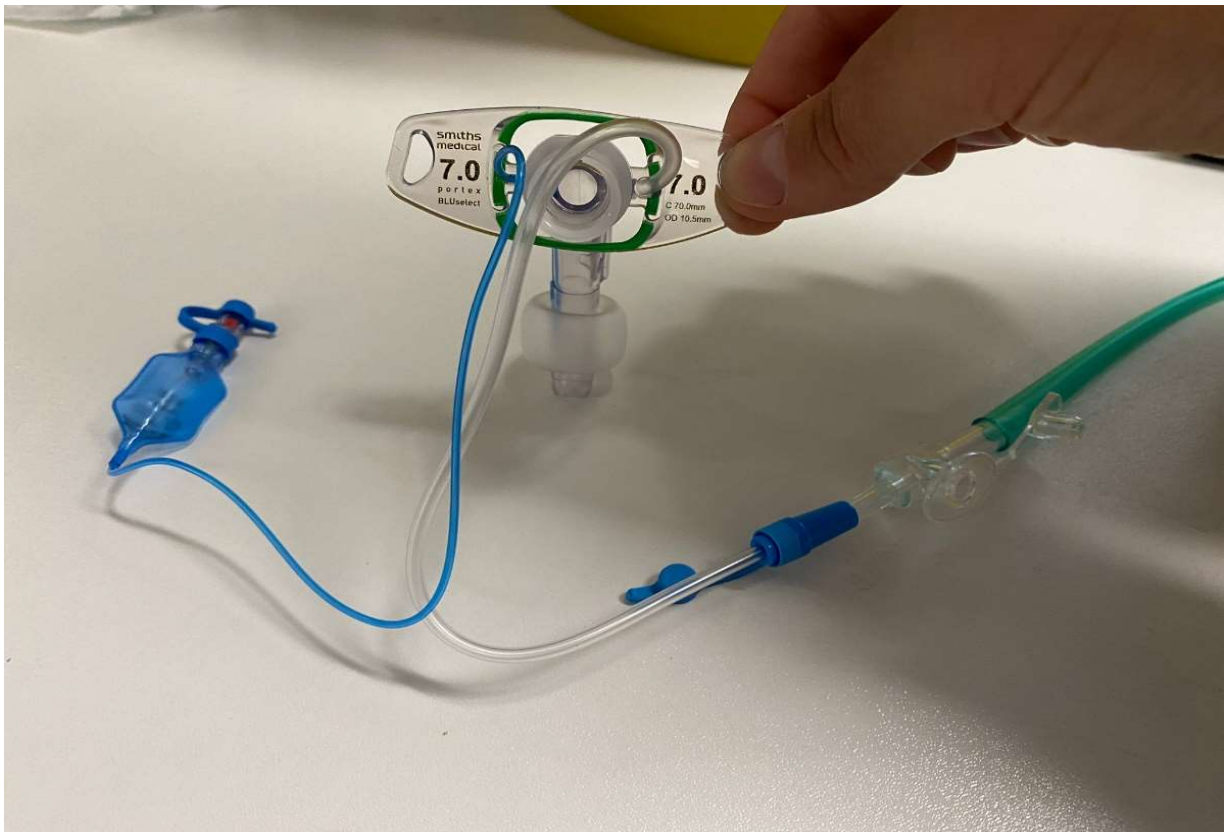


Figure 5

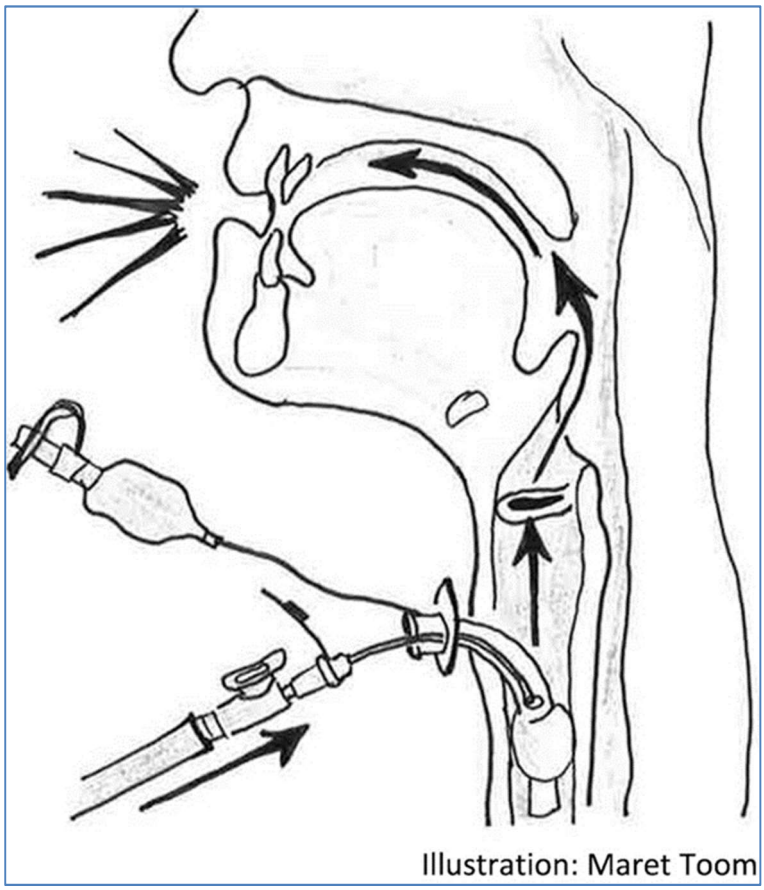
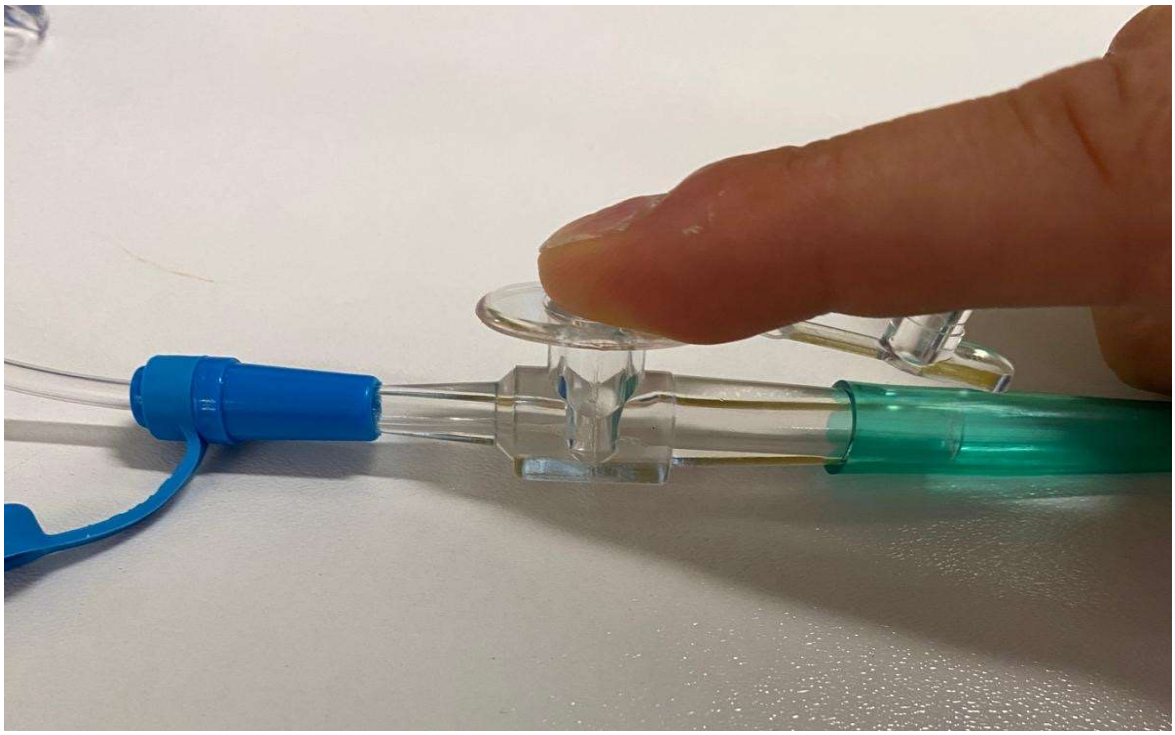


Figure 6



Potential Side-effects/ risks:
The following side-effects have been reported with the use of ACV:

- Discomfort
- Excessive oral secretions
- Stomal air leak
- Laryngeal drying
- Gagging
- Nausea
- Potential surgical emphysema (potential risk only, no documented incidences)

In the event of any of the above, ACV should be discontinued and discussed with multi-disciplinary team prior to further use.

Documentation:
It is essential that use of ACV is appropriately documented including outcome, ongoing tracheostomy weaning plan or reason for discontinuation of ACV. This will ensure MDT awareness of plan and ensure overall success of the weaning plan.

6. ARCHIVING ARRANGEMENTS

The original of this SOP, will remain with the author Ruth Wood & Laura Gilham. An electronic copy will be maintained on the Trust intranet, P – Policies – ACV. Archived electronic copies will be stored on the Trust's “archived policies” shared drive and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years.

7. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE/ GUIDELINE

To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Datix in relation to ACV	*Number and severity of issue*	*Helen Marshall (matron for ICU)* *Nicholas Love (clinical lead for ICU)* *Rachel Nuth (Inpatients therapy lead)* *Claire Hakin (SLT Professional Lead)*

8. REFERENCES

National Tracheostomy safety project: https://tracheostomy.org.uk/healthcare-staff/vocalisation/acv-above-cuff-vocalisation

Mills, Micho, and King (2022). Evidence for Above Cuff Vocalization in Patients with a Tracheostomy: A Systematic Review. Laryngoscope. 2022 Mar;132(3):600-611

Mills, Michou an Bellamy (2024) Worth a try or a last resort: Healthcare professionals' experiences and opinions of above cuff vocalisation. Int J Lang Commun Disord. 2024 May-Jun;59(3):932-946.

Mills, Cuthbertso and Michou (2023) What's new in reducing the impact of tracheostomy on communication and swallowing in the ICU. Intensive Care Medicine 49

Petosic, Viravong, and Martin (2021) Above cuff vocalisation (ACV): A scoping review. Acta Anaesthesiol Scand. 2021 Jan;65(1):15-25

APPENDIX 1: COMMUNICATION PLAN

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the SOP	All staff that work on ICU
The key changes if a revised document	New document
The key objectives	Safe and consistent use of ACV.
How new staff will be made aware of the procedure and manager action	ICU governance, ICU newsletter.
Specific Issues to be raised with staff	Contraindications section Procedure
Training available to staff	Training provided by consultants trained in ACV, Laura Gilham, Claire Hakin, Lisa Bashford, Ruth Wood
Any other requirements	
Issues following Equality Impact Assessment (if any)	NA
Location of hard / electronic copy of the document etc.	g: ICU/ACV